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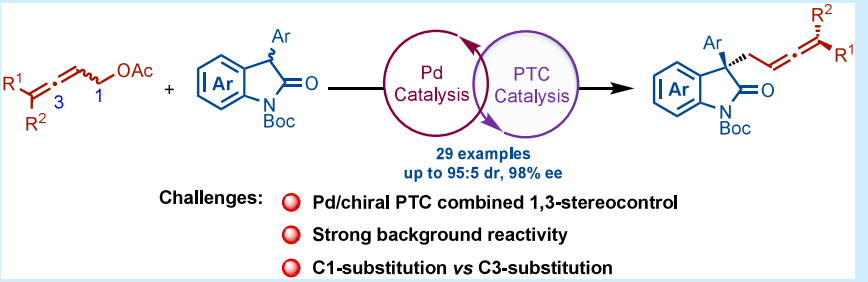
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1,3-Stereocontrol in Stereoconvergent Allenylic Alkylation Enabled by Palladium and Chiral Phase-Transfer Cooperative Catalysis

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ABSTRACT: A stereoconvergent allenylic alkylation of allenyl acetates with oxindoles has been established by palladium and chiral phase-transfer cooperative catalysis, allowing for an efficient construction of 1,3-nonadjacent stereogenic elements featuring both central and axial chirality. The noncovalent interactions of PTC create a chiral environment for nucleophilic attack on η^3-allylpalladium intermediates. Mechanistic investigations reveal that the Pd/PTC synergistic action and matched chirality are both critical for stereocontrol, while PTC primarily determines the stereoselectivity. Notably, this work represents the first successful implementation of Pd/chiral PTC dual catalysis for 1,3-nonadjacent stereocontrol, thereby enriching the synthetic toolbox for constructing challenging remote stereocenters.

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Asymmetric catalysis stands at the forefront of modern organic synthesis, enabling the stereocontrolled assembly of enantioenriched molecules vital to pharmaceuticals, agrochemicals, and functional materials.[1] The prevalence of multiple stereogenic elements in bioactive molecules[2] has made stereoselective construction a long-standing research focus in organic synthesis.[3] Despite remarkable advances in vicinal stereocenter construction, the precise control of nonadjacent stereocenters remains a persistent challenge.[4] Three interrelated factors contribute to this synthetic challenge:[5] (i) the flexible transition state emerges as a critical barrier to reliable remote stereocontrol; (ii) asymmetric induction efficiency declines with greater stereocenter separation; and (iii) nonadjacent stereocenter formation produces ≥4 stereoisomers, increasing synthetic complexity.

Chiral allenes represent a privileged class of structural motifs found in natural products, bioactive molecules and functional materials.[6] Consequently, the design of efficient synthetic routes to chiral allenes has become a major objective in organic synthesis, and significant advances have recently been achieved in the field of constructing single allenyl axial chirality.[7] Transition-metal-catalyzed asymmetric allenylic alkylation represents a powerful strategy for the stereoselective construction of enantioenriched allene derivatives with precise stereocontrol.[7d,8,9] Notably, allenyl axial chirality with vicinal central chirality could be controllably installed using transition-metal-catalyzed.[8g,10] However, the simultaneous control of remote stereocenters featuring both allenyl axial chirality and central chirality remains limited,[8i,j,9] despite the fact that such scaffolds play a significant role in medicinal chemistry and synthetic chemistry.[11] In 2023, two distinct Pd/chiral amine dual catalyst systems were independently developed by Mi and Luo[9e] and Li and Shao[9b] for allylic allenylation reactions of β-ketocarbonyl and aldehyde, respectively, enabling the 1,3-nonadjacent stereocenters with allenyl axial chirality and central chirality to be stereoselectively controlled. Subsequently, Wen and Guo described a Pd/chiral aldehyde cooperatively catalyzed asymmetric allenylic alkylation of unprotected amino acid esters, providing an efficient access to these valuable scaffolds (Scheme 1a).[9f] Beyond metal/organocatalytic approaches, bimetallic catalysis has proven highly effective for precise stereocontrol in constructing these scaffolds, as demonstrated by Zhang and Ma[8j,9g] and He[9h] and Lin.[9a]

Chiral PTCs are important type of organocatalysts and have grown to be indispensable tools for asymmetric synthesis,

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Scheme 1. Strategies for the Formation of Multiple Stereocenters via Transition Metal/Organo Dual Catalysis

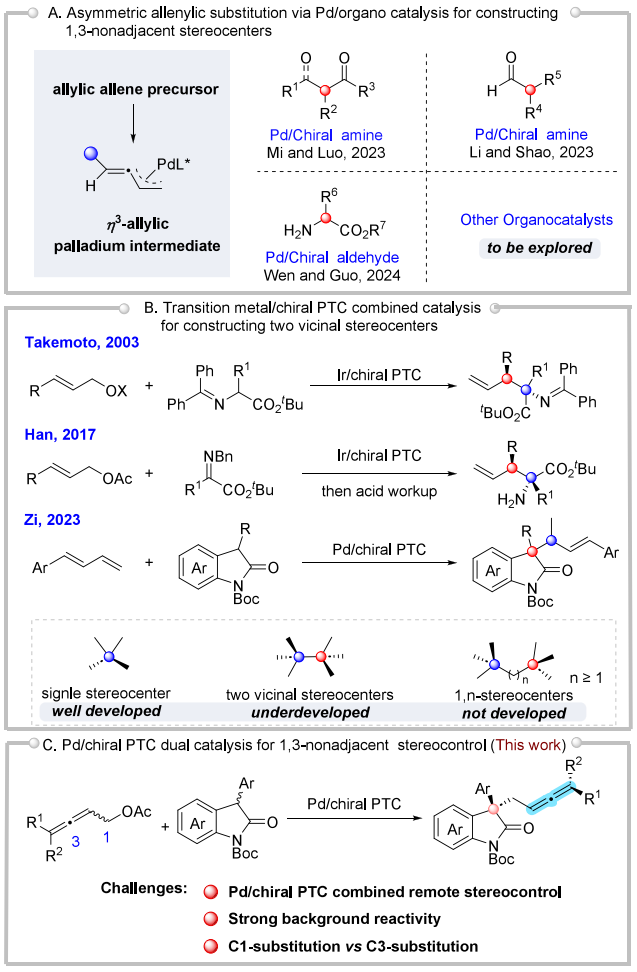
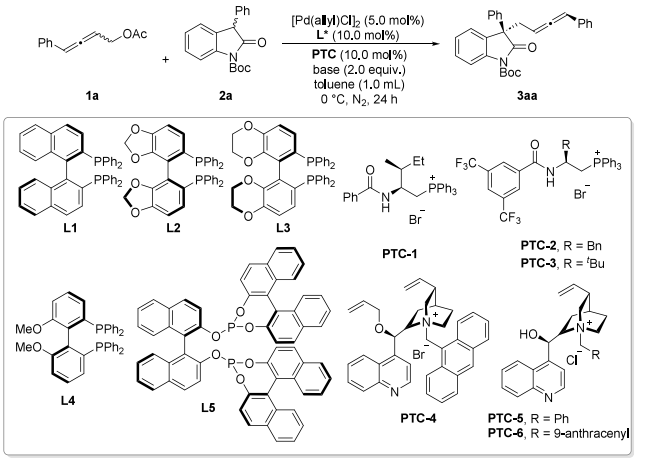


Table 1. Optimization of Reaction Conditions[a]



entry	L*	PTC	base	yield (%)[b]	dr[c]	ee (%)[d]
1	**L1**	**PTC-1**	KOAc	65	65:35	44
2	**L2**	**PTC-1**	KOAc	80	63:37	68
3	**L3**	**PTC-1**	KOAc	84	64:36	58
4	**L4**	**PTC-1**	KOAc	93	66:34	66
5	**L5**	**PTC-1**	KOAc	27	70:30	72
6	**L5**	**PTC-2**	KOAc	34	62:38	18
7	**L5**	**PTC-3**	KOAc	51	84:16	94
8	**L5**	**PTC-4**	KOAc	12	83:17	11
9	**L5**	**PTC-5**	KOAc	33	60:40	56
10	**L5**	**PTC-6**	KOAc	<10	67:33	70
11	**L5**	**PTC-3**	Na_2CO_3	45	86:14	92
12	**L5**	**PTC-3**	K_2CO_3	57	85:15	93
13	**L5**	**PTC-3**	K_3PO_4	51	86:14	89
14	**L5**	**PTC-3**	NaOH	65	93:7	95
15	**L5**	**PTC-3**	KOH	55	67:33	58
16[e]	**L5**	**PTC-3**	NaOH	74	93:7	95
17[e,f]	**L5**	**PTC-3**	NaOH	80	95:5	96

[a]Reaction conditions: **1a** (0.09 mmol, 1.5 equiv), **2a** (0.06 mmol, 1.0 equiv), $[Pd(allyl)Cl]_2$ (5.0 mol %), ligand (10.0 mol %), **PTC** (10.0 mol %), base (2.0 equiv), and toluene (1.0 mL) at 0 °C for 24 h under a nitrogen atmosphere. [b]Isolated yields were based on **2a**. [c]The diastereomeric ratio (dr) of **3aa** was determined by ^{1}H NMR spectroscopy. [d]The enantiomeric excess (ee) of the major diastereomer of **3aa** was determined by chiral HPLC. [e]2.5 equiv of NaOH was used. [f]$[Pd(allyl)Cl]_2$ (8.0 mol %), **L5** (16.0 mol %) and **PTC-3** (16.0 mol %) for 36 h were used.

uniquely enabling stereocontrol of anionic nucleophiles via well-defined ion-pair interactions.[12] Notably, this system faces heightened background reactivity relative to classical organocatalysts. Additionally, PTC systems feature a single activation site, specifically a chiral cationic center that mediates ionic interactions, restricting the simultaneous activation of both reaction partners. This limitation prompted developing PTC-based dual catalytic systems with secondary activation modes, significantly broadening synthetic applications.[13] Since the Pd/chiral PTC-catalyzed asymmetric allylic alkylation of Schiff bases reported by Gong[14a] and Takemoto,[14b] transition metal/chiral PTC combined catalysis has emerged as a highly promising strategy in asymmetric catalysis.[15] However, such catalytic systems have rarely been applied to simultaneously and stereoselectively construct 1,3-nonadjacent stereocenters (Scheme 1b). We herein report a Pd/chiral PTC-catalyzed enantioselective allenylic alkylation, simultaneously creating allenyl axial chirality and all-carbon quaternary stereocenters with excellent enantioselectivity and diastereoselectivity (Scheme 1c).

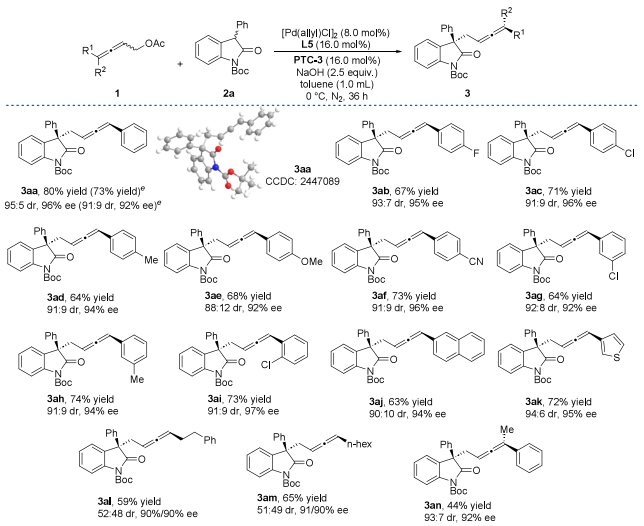
We initiated a systematic study of the Pd/chiral PTC-catalyzed asymmetric allenylic alkylation using racemic allenylic acetate **1a** and oxindole **2a** as model substrates (Table 1). Several commercially available ligands (**L1**−**L4**) were evaluated in the presence of an L-isoleucine-derived quaternary phosphonium salt (**PTC-1**). While these systems could deliver the desired allylic allenylation product **3aa**, both the enantioselectivity and diastereoselectivity proved unsatisfactory (entries 1−4). The observed suboptimal stereocontrol may be attributed to competing background reactivity mediated by a single Pd/ligand catalytic cycle. Inspired by findings from Takemoto[14b] and Zi,[15f] the deliberate use of electronically more defficeint phosphite ligand **L5** would be able to slow down the competing monocatalytic process. Remarkably, this approach not only substantially enhanced enantioselectivity but also significantly improved diastereoselectivity, albeit at the expense of the reaction yield (entry 5). Subsequently, screening of various phase-transfer catalysts (PTCs) identified **PTC-3** as the optimal, demonstrating remarkable enhancement in the yield, diastereoselectivity, and enantioselectivity. These results strongly suggest a synergistic catalytic effect between ligands **L5** and **PTC-3**, as evidenced by comparative evaluation of different PTC systems (entry 7). Base optimization (entries 11−15) identified NaOH

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as critical for maximizing both reactivity and selectivity in the **L5**/**PTC-3** system, achieving peak performance (entry 14). Systematic optimization of other reaction parameters, including catalyst loading, base stoichiometry, and reaction time, found that increased yields and slightly improved stereoselectivity were obtained upon using slightly increased amounts of the palladium catalyst and chiral PTC (entries 16 and 17).

With the optimized reaction conditions in hand, we next explored the substrate generality of the Pd/chiral PTC-catalyzed asymmetric allenyl alkylation reaction. We first examined the scope of allenyl acetates **1** (Scheme 2). The

Scheme 2. Scope of the Allenylic Acetates[a−d]

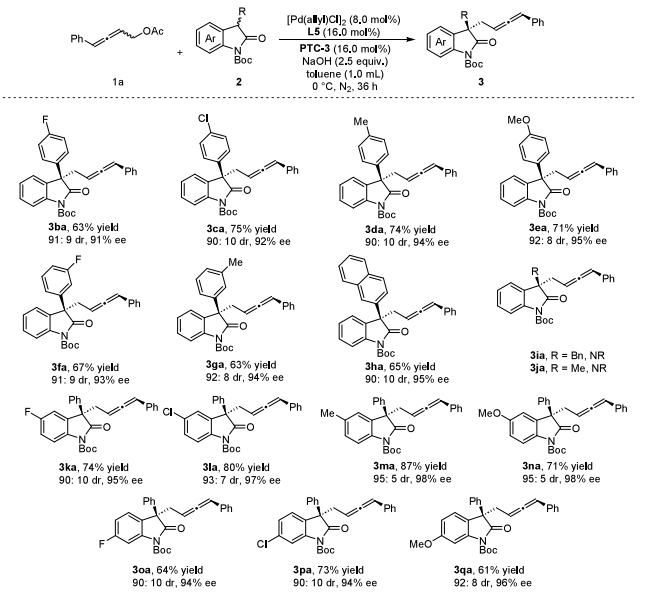


[a]Reaction conditions: **1** (0.09 mmol, 1.5 equiv), **2a** (0.06 mmol, 1.0 equiv), $[Pd(allyl)Cl]_2$ (8.0 mol %), **L5** (16.0 mol %), **PTC-3** (16.0 mol %), NaOH (2.5 equiv), and toluene (1.0 mL) at 0 °C for 36 h under a nitrogen atmosphere. [b]Isolated yields were based on **2a**. [c]The diastereomeric ratio (dr) of **3** was determined by ^{1}H NMR spectroscopy. [d]The enantiomeric excess (ee) of the major diastereomer of **3** was determined by chiral HPLC. [e]Performed on 1.0 mmol scale.

presence of a fluoro, chloro, and methyl group on the benzene ring at the *para*- position could smoothly provide the products **3ab**−**3ad** in good yields and excellent stereoselectivities (dr >91:9, >94% ee). However, a highly electron-donating substituent is detrimental to the reaction rate and stereoselectivity as exemplified by the 4-OMe-substituted substrate, which delivered the product **3ae** in moderate yield and eroded diastereoselectivity (dr 88:12). Gratifyingly, the good isolated yield and stereoselectivities were secured for target product **3af** when a substrate possessing an electron-withdrawing group (−CN) at the *para*- position of the benzene ring was employed. The substitution pattern of the aromatic ring significantly influenced the outcomes. As a result, the substituent (−Cl, −Me) at either the *meta* or *ortho* position underwent reaction smoothly, yielding products **3ag**−**3ai** with good stereoselectivity (dr >91:9, >92% ee). A 2-naphthyl-substituted allenic acetate (**2j**) proved compatible, furnishing **3aj** in 63% yield with 90:10 dr and 94% ee. The protocol also accommodated the heteroaryl allenic substrates. For example, the thiophene-functionalized substrate participated in the reaction to afford **3ak** in 70% yield with excellent stereoselectivity (dr 94:6, 95% ee). For alkyl substituted substrates, linear alkyl allenic acetates were reactive substrates, leading to **3al** and **3am** in acceptable yields (59−65%) and high enantioselectivities (ee >90%), but diastereoselectivity was rather poor. Trisubstituted allenyl acetate **2n** could participate in the reaction with excellent enantioselectivity (>92% ee), but gave moderate yield. The absolute configuration of **3aa** was unambiguously determined by single-crystal X-ray analysis. Notably, the reaction scalability was confirmed through a 1.0 mmol scale preparation of **3aa**.

We next evaluated the reaction scope with respect to oxindoles **2** (Scheme 3). The catalytic system demonstrated a

Scheme 3. Scope of the Oxindoles[a−d]



[a]Reaction conditions: **1a** (0.09 mmol, 1.5 equiv), **2** (0.06 mmol, 1.0 equiv), $[Pd(allyl)Cl]_2$ (8.0 mol %), **L5** (16.0 mol %), **PTC-3** (16.0 mol %), NaOH (2.5 equiv), and toluene (1.0 mL) at 0 °C for 36 h under a nitrogen atmosphere. [b]Isolated yields were based on **2**. [c]The diastereomeric ratio (dr) of **3** was determined by ^{1}H NMR spectroscopy. [d]The enantiomeric excess (ee) of the major diastereomer of **3** was determined by chiral HPLC.

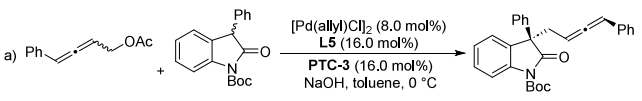
broad tolerance for diverse electronic perturbations at the *para*-position of the C3-aryl group. Both electron-withdrawing (−F, −Cl) and electron-donating (−Me, −OMe) substituents participated in the reaction, smoothly, affording the products **3ba**−**3ea** in good yields and stereoselectivities (dr >90:10, >91% ee). The method proved equally effective for meta-substituted C3-aryl oxindoles. Either chloro or methyl substituted substrate delivered the products **3fa** and **3ga** in good yields and excellent stereoselectivity (dr >91:9, ee >93%). Expansion to larger aromatic systems was also successful, as shown in the case with the 2-naphthyl substrate, which furnished **3ha** in 65% yield with 90:10 dr and 95% ee. However, the system showed clear limitations with alkyl substituents at C3. Neither methyl nor benzyl substituted oxindoles generated target products **3ia** and **3ja**, likely due to the diminished acidity of the C3−H bond in alkyl-substituted substrates. The protocol exhibited excellent compatibility with C5-substituted oxindoles. Halogenated substrates (**3ka** and **3la**) underwent the reaction with excellent stereocontrol (dr >90:10, >95% ee), while electron-rich variants (**3ma** and **3na**)

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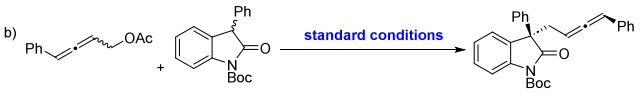
achieved even better results (95:5 dr, 98% ee). Similarly, C6-substituted oxindoles bearing a fluoro (**3oa**), chloro (**3pa**), or methoxy (**3qa**) group performed admirably, providing products with good enantioselectivity (>94% ee).

To elucidate the reaction pathway, a series of mechanistic investigations were conducted (Scheme 4). First, some related

Scheme 4. Mechanistic Investigations[a]



entry	variation from the standard conditions	yield (%)	dr	ee (%)
1	none	80	95:5	96
2	w/o **L5**	n.d.	-	-
3	w/o **PTC-3**	75	72:28	6
4	with *ent*-**L5**	78	86:14	96
5	with *ent*-**PTC-3**	85	82:18	-96



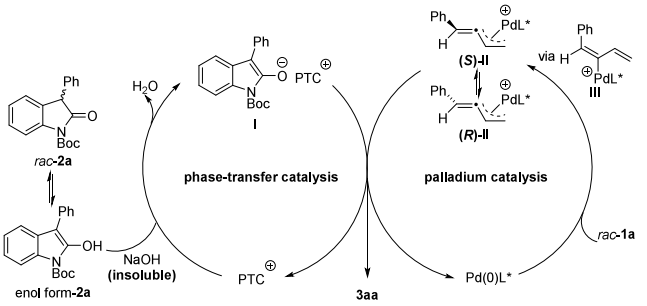
entry	variation	yield (%)	dr	ee of **3aa** (%)
1	using (*R*)-**1a**	76	95:5	96
2	using (*S*)-**1a**	72	95:5	96

[a]The ee for both (*R*)-**1a** and (*S*)-**1a** was 93%.

control experiments were performed (Scheme 4a). No product formation occurred in the absence of the chiral ligand (Scheme 4a, entry 2), confirming its essential role in this transformation. Omitting chiral PTC yielded almost racemic **3aa** (ee <10%) with poor diastereocontrol (Scheme 4a, entry 3). The result underscores the dual role of chiral PTC in both phase-transfer activation and enantioselective nucleophile delivery. Configuration inversion of either **L5** or **PTC-3** significantly eroded diastereoselectivity (Scheme 4a, entries 4 and 5), demonstrating that chiral complementarity between the phosphine ligand and phase-transfer catalyst is essential for optimal stereocontrol in this dual catalytic system. This matched/mismatched phenomenon suggests cooperative transition-state organization involving both chiral elements. Notably, *ent*-**PTC-3** generated products with inverted enantiomeric excess (ee) values. These observations collectively demonstrate that the phase tranfer catalyst serves as the primary determinant of the stereoselectivity in this transformation. To probe potential kinetic resolution effects, the reaction was performed by using enantiomerically pure (*R*)- and (*S*)-**1a** as substrates (Scheme 4b). Intriguingly, both enantiomers efficiently underwent the asymmetric alkylation, affording the products with the same configuration, in similar yields and with identical enantioselectivity. This observation suggests that the quick racemization allows both substrate enantiomers of **1a** to be funneled into a single product stereoisomer, leading to the enantioconvergent nature.

The proposed mechanism involves two catalytic cycles, as illustrated in Scheme 5. In the PTC cycle, PTC+ transfers the base OH- from the solid base to the organic phase. The enol form of 2a undergoes deprotonation by the base, generating a tightly associated nucleophilic ion pair (I) with PTC$^+$, along

Scheme 5. Proposed Mechanism



with one equivalent of water. In the pal-ladium catalysis cycle, oxidative addition of Pd(0) to ra-cemic allene 1a to form both η^3-allylpalladium interme-diate (R)-II and (S)-II, which rapidly equilibrates via η^1-allylpalladium intermediate III. The (S)-II intermediate is kinetically favored over its (R)-II counterpart for nucleo-philic attack by the chiral ion pair I. This preferential re-activity pathway enables the highly stereo-selective in-termolecular asymmetric remote alkenylation, ultimately leading to the formation of product 3aa with excellent stereocontrol. In summary, we have established a stereoconvergent enantioselective allenylic alkylation of oxindoles with allenyl acetates enabled by cooperative catalysis of the chiral palladium complex and phase-transfer catalyst (PTC), leading to the highly stereoselective construction of 1,3-nonadjacent stereogenic elements featuring both central and axial chirality. The key to successful stereocontrol is the use of PTC's noncovalent interactions to establish a chiral environment for the stereoselective nucleophilic addition to the η^3-allylpalladium intermediates. Mechanistic studies reveal that the synergistic cooperation between the palladium catalyst and PTC, along with their matched chirality, plays a critical role in enabling excellent reaction performance (Scheme 5). This work significantly expands the scope of PTC applications in metal/organo cocatalysis. We anticipate that this strategy will provide a general platform for the stereoselective construction of remotely functionalized chiral centers in more challenging contexts.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02416.

Experimental procedures, characterization data and Copies of ^{1}H, ^{13}C NMR and ^{19}F NMR spectra for all new compounds and crystallographic data for compound **3aa**. (PDF)

Accession Codes

Deposition Number 2447089 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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